TM Forum Specification

Open Gateway Product Usage User Guide

TMF937

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INTRODUCTION

This document is the User Guide for Product Usage API within the framework of the GSMA Open Gateway initiative, enabling automated usage exchange and settlement between Operators and Channel Partners.

This API is inspired by TMF635 Product Usage API v5.0 for Open Gateway context.

The GSMA Open Gateway initiative defines standardized Service APIs and Service Management APIs, complemented by a common set of Operate APIs that provide cross-cutting commercial and operational capabilities. These Operate APIs define homogeneous integration mechanisms between Operators and Channel Partners to ensure interoperability across networks and markets.

By defining clear and consistent API requirements, this document supports alignment with industry best practices and facilitates communication among stakeholders in the Open Gateway ecosystem. This version focuses on the Usage Consumption Reporting function within the Operate API suite, while future versions may include other Channel Partner integration capabilities such as service assurance, monitoring, and settlement automation.

OVERVIEW OF GSMA OPEN GATEWAY AND CHANNEL PARTNER MODEL

Open Gateway is a GSMA-led initiative in the telecommunications sector that turns communication networks into platforms by exposing telco capabilities through industry-standard APIs on a global scale under the CAMARA open-source project. CAMARA is hosted by the Linux Foundation and provides the technical framework for these standardized APIs.

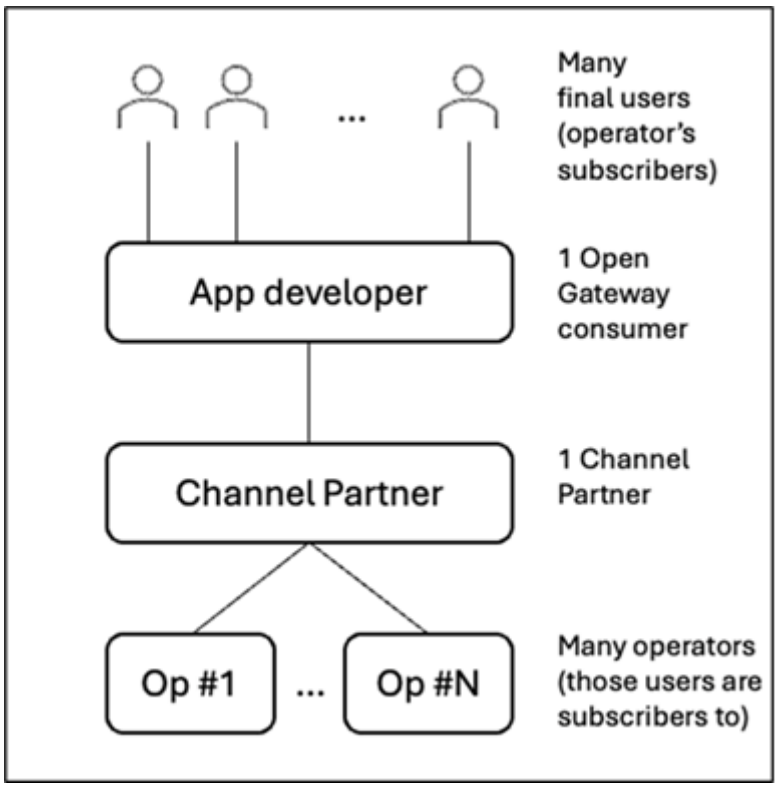
Channel Partners are key go-to-market players for Operators that want to sell APIs to many Developers who do not want to integrate individually with each Operator. By aggregating multiple Operators' APIs into a single Operator-agnostic interface, Channel Partners can offer these capabilities to a wide community of application Developers.

Channel Partner model overview

Open Gateway Channel Partner model is composed of different players:

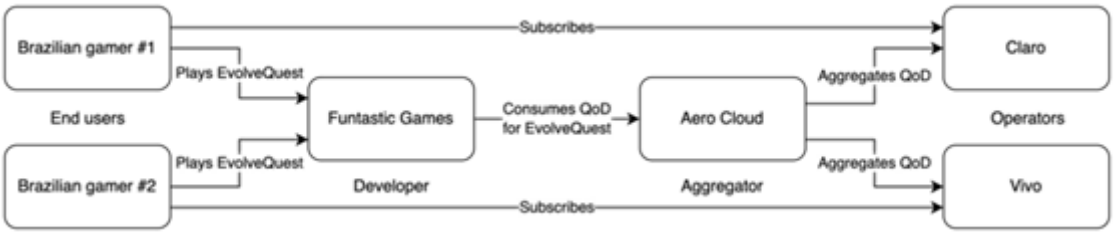
- End Users are both subscribers to Operators and users of the application published by the Developers.
- Developers (Application Owner Organization) build and own the applications that consume APIs from the Operators to provide End Users with enhanced features or enable new use cases.
- Channel Partners aggregate Operators' CAMARA standardised APIs to build Open Services and implement Operator end-point routing based on final user identification on the networks.
- Operators expose telco capabilities through CAMARA standardised APIs and could partner with Channel Partners to enable the Services that they offer to Developers.

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An illustrative scenario can involve a game publisher using a cloud provider that acts as a Channel Partner and leverages Open Gateway products such as QoD to enhance mobile gaming experiences. The ability of Operators and Channel Partners to form agreements directly affects which subscribers can access those enhanced services.

Two gamers using the same game but different mobile Operators are mapped to different Operator–Channel Partner relationships, as shown in the relational diagram.



Example Interaction between players on the model

Operate APIs

As indicated, the Channel Partner indirect model enables a single channel for tenant applications to gain access to capabilities from multiple Operators, without the need for the customer to set up a contractual relationship with each of them. It is clear the need to provide certain additional functionalities from the Operators to the Channel Partner to enable the commercial model.

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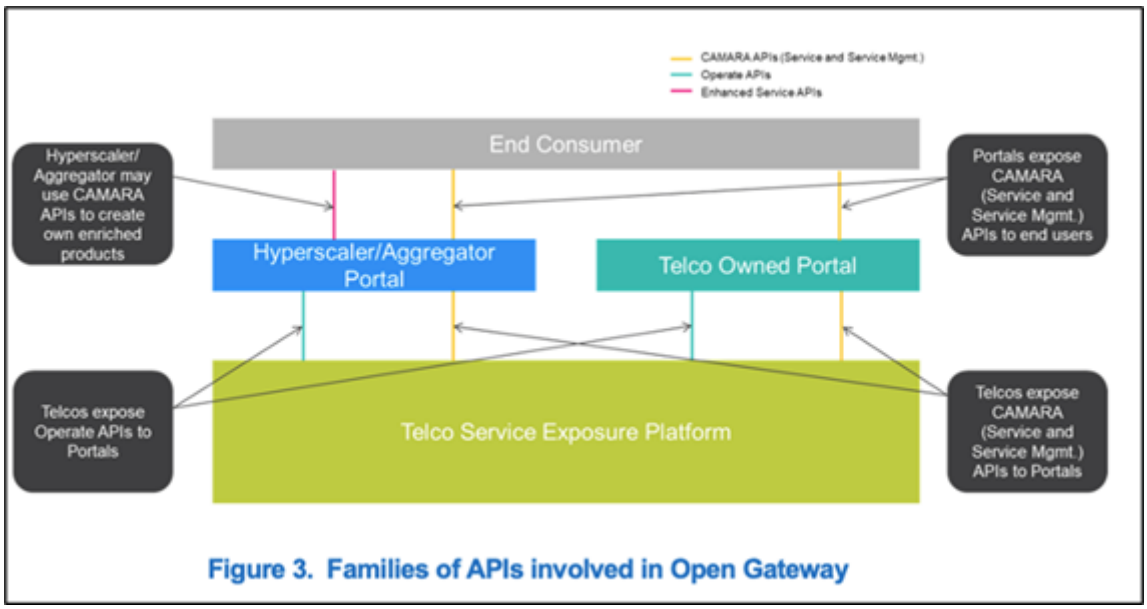


Figure 3. Families of APIs involved in Open Gateway

Therefore, in addition to the CAMARA APIs, the Operate APIs are proposed for offering programmable access to Operation, Administration and Management (OAM) capabilities to facilitate the integration of the Open Gateway NaaS Platform with Channel Partner. These APIs provide all the transversal/non-service specific functionalities that are required to make a commercial product out of the Service APIs and Service Management APIs, making them operable and monetizable. Examples of functions provided by the Operate APIs include ordering, usage consumption reporting, monitoring, trouble ticketing, accounting, billing, payment, etc., but also API discovery, user registration and consent management, etc.

Scope

This specification focuses on the Usage Consumption Reporting function within the Operate API suite. Its purpose is to define the requirements and aggregated file structure for exchanging ProductUsage data between Operators and Channel Partners. This capability enables automation of the usage collection, reporting, and settlement process, which is essential for monetizing Open Gateway APIs in both the Wholesale Reseller and Marketplace Aggregator models. Operators record each invocation of CAMARA APIs and transform those invocations into ProductUsage records, following the TM Forum Open API and SID information models. These ProductUsage records are then aggregated and shared in bulk with Channel Partners via the Usage Operate API.

Commercial Models Supported

Model	Description
Wholesale Reseller	The Channel Partner manages the service, defines its own end-user products and pricing, and receives usage data from the Operator for settlement.
Marketplace Aggregator	The Operator defines the product and pricing; the Channel Partner resells it and requires usage data that already includes rated information for end-user billing.

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In both models, the basis of exchange is a structured set of ProductUsage records containing sufficient attributes to enable pricing, billing, and reconciliation. These records are equivalent to CDRs (Call Detail Records) or EDRs (Event Detail Records). The difference lies in whether pricing details are included (Marketplace) or applied later (Wholesale), i.e. rated or non-rated.

Key Differentiator – What's New for Usage

Compared with other Operate APIs such as Ordering, the Product Usage Operate API introduces several specific capabilities.

- A standardized ProductUsage data model aligned with TMF635 v5.0 Product Usage API.
- An embedded ExportJob mechanism for asynchronous bulk data transfer, aligned with TMF635 evolution.
- Defined file and metadata structures using ZIP, JSON and signatures for auditability and interoperability.
- Support for mutual TLS to ensure trusted exchange between parties.
- Compatibility with both Marketplace and Wholesale commercial models.

These differentiators make the Product Usage Operate API a bridge between the commercial product catalog and operational monetization layer. Once APIs are defined and offered in the Operator's catalog, the Product Usage Operate API provides the mechanism for exchanging measured usage data that directly relates to those catalog entries and pricing models.

Catalog Alignment

The Product Usage Operate API complements the TMF936 Open Gateway Product Catalog API by defining how usage of catalogued CAMARA APIs is measured, recorded, and shared between the Operator and Channel Partner. In other words, while the catalog defines what is being offered (e.g., QoD, SIM Swap, Number Verification) and under what commercial terms, the Product Usage Operate API defines how the consumption of those offers is operationally reported and monetized. The two together — catalog definition and usage reporting — form the foundation for automated settlement and revenue assurance across the Open Gateway ecosystem.

The catalog for Operator-exposed APIs includes:

- A CAMARA standardized API available on the Operator's exposure platform.
- Defined API scopes (e.g., check-sim-swap, verify-number) and corresponding usage characteristics.
- Supported charging models (one-shot, recurring, usage-based).
- Applicable purposes and constraints (privacy, SLA, or policy).

The Product Usage Operate API extends this catalog context by standardizing the ProductUsage exchange for each CAMARA API and aligning usage characteristics with the corresponding ProductUsageSpecification that can be defined in TMF767 Product Usage Catalog API.

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SAMPLE USE CASES

Usage refers to the specific instances of Open Gateway API service consumption by end users or applications. These events capture critical data about how, when and by whom a service was used and the products used in the Operator to authorize that service usage. They are the foundational data that drives billing, revenue sharing, and operational insights. Its meticulous tracking and transparent reporting are vital to the integrity and efficiency of modern telecom ecosystems.

In the context of the GSMA Playbook, the Operate API related to Usage are in the context of exchanging data between the Operator and the Channel Partner but could also be considered in the context of a direct relationship between Operator and Developer.

The exchange of information between an Operator and a Channel Partner can be done in a Wholesale Reseller or a Marketplace Aggregator Model:

Wholesale Reseller Model	Marketplace Aggregator Model
Services fully managed by the Channel Partner Channel Partner enters an agreement with Operators for their services Channel Partner manages the end user products, performs pricing and invoices developer Operator provides the underlying service and tracks the usage Operator provides the Channel Partner with the usage to support them in billing processes and for reconciliation. Operator provides the service at a wholesale rate and invoices the Channel Partner Channel Partner deduct a percentage fee to the amount paid by the Developer and make an account deposit to the Operator for the remaining amount	Operators upload their products to the Channel Partner's Marketplace and enter an agreement for their services Channel Partners provide a unified interface for developers to ensure a coherent user experience Developers purchase products from the Marketplace Operator provides the underlying service and tracks the usage The usage is priced according to the Operator's product definition Operator provides the Channel Partner with the priced usage required to support them in their invoicing process Operator keeps track of their revenue share based on the priced usage and invoices Channel Partner

For both models, the basis of the information exchange is sufficient details of the use of the APIs to be able to price the event based on required attributes according to the product definition. The key difference between the two types is that the Marketplace Aggregator Model includes how the Operator charged for the CAMARA API usage events based on the product definition. The requirements in this chapter focus on the requirements on the usage consumption exchange between the Operator and Channel Partner for both models above.

GSMA Playbook requirements

The below is a summarization of the requirements:

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#	Summary	Comment
GSMA0	Definition of a unified way to report usage information of the Open Gateway services, so that aggregators/Channel Partners can make it available to the developers accordingly. Information related to consumption of Service APIs is managed by the Operators who are the ones who have an end-to-end view of the services and network resources. This information is exposed towards aggregators/Channel Partners for consolidation and billing management.	
GSMA1	Service Charging process will be performed based on the consumption of the Open Gateway Products and will involve the following: Charging Calculation: charging calculation will be implemented by the Operator based on the consumption of the specific API Operations. The calculation will be performed based on billable concepts that could be different depending on the Open Gateway Product. Examples of these billable concepts are contracting, exchange, time duration, data volume or characteristics (modifiers). The information associated to these billable concepts will be managed by Operators, as they have the control of details of the Open Gateway Product execution. Operators will implement the required calculation process (guiding) to generate CDRs (Call Detail Records) or EDRs (Event Detail Records) which include the associated data and rating to calculate price.	In the context of TM Forum, the CDR/EDR is not an entity but the CDR/EDR concept corresponds to the Usage entity. The exchange of information by an Operator to an external party is done on Product Usage level based on the underlying Service Usage. The focus for the Operate API will be to exchange the Product Usage information but to do so, there needs to be an understanding of what elements of the Service Usage to bring in and surface as well as adding Product Usage attributes, for example pricing elements.

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#	Summary	Comment
GSMA2	Share usage information from Operator to Channel Partner to be able to generate the required billable information. Operators are responsible for implementing the mechanism to share this information with the Channel Partners. Charging information could be shared using two different approaches. * Operator exposes information to Channel Partner: the charging information will be exposed by the Operator using Operate APIs that the Channel Partner can invoke. Operators should implement these APIs and make them available to Channel Partners that will implement the associated process to manage charging information. * Operator loads the information onto the Channel Partner: the charging information will be loaded to the Channel Partner systems by invoking their proprietary APIs. Channel Partners usually implement these types of APIs as part of their standard service. Operators need to implement the associated process to load the required charging information.	In the context of Operate API Usage definition, the first approach is applicable. However, the content of the data exchanged in both cases should be similar for the same type of service. The pull mechanism, frequency etc. will need to be considered due to the expected high volumes of data to be exchanged.
GSMA3	Billable concepts are defined as a feature associated with the Service that can be measured and accounted for, and to which a value is assigned. The billing process will use these values to charge for the Service.	See the initial list of billable concepts for more details

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#	Summary	Comment
GSMA4	Charging aggregation levels. * Depending on the business model, different concepts will need to be shared with the Developer so they can bill their customers. To do so, charging information should be aggregated to provide the required detail to implement the associated commercial models. * In general it will be required to generate an “Aggregation at Application Developer Services level” where all the concepts associated with a specific application of a Developer will be aggregated, allowing for a detailed breakdown of the entire consumption of a Service associated with this Developer and application (i.e. the time associated with the Quality of Data (QoD) Service for Developer 1 AR application). Operate APIs * The Operate APIs involved in the charging and billing flows are for Consumption reporting **Information related to consumption of Service APIs is managed by Operators, who are the ones having an end-to-end view of the services and network resources.** Definition of a unified way to report usage information of the Open Gateway API Products, so that Channel Partners can make it available to Developers accordingly.	In the first version of the Operate API for Usage, the focus is to exchange the individual usage records as this forms the basis for any future enhancements.

Below is the initial list of the billable concepts with more details:

Billable Concepts	Description	Explanation	Examples
Contracting	Billable concept associated with permissions and availability to execute a specific Service/API.	Recurring/activation fee associated with the use of Services.	API registration

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Billable Concepts	Description	Explanation	Examples
Exchange	Billable concept that will measure the calls or responses associated to a Service/API, or the events received based on a subscription.	Fee associated with the number of API calls/responses performed.	API call (including subscription activation); API notification
Time	Billable concept that will measure the time the Service/API has been active.	Fee associated with active minutes runtime of the Service.	Session duration; subscription duration
Data	Billable concept that will measure the traffic the Service/API has consumed.	Fees associated with the volume of data consumed by the Service.	Session consumption
Characteristic (modifier)	This modifier can affect the real value of any of the previous concepts, including contracting.	Characteristics associated to the particular level of service or response.	Number of parameters, location accuracy, data age, quality profile, share of amount in direct carrier billing

Product Usage Requirements

Each Open Gateway Service API needs its own definition of Product Usage Specification for Wholesale and for Marketplace Aggregator model to exchange information for the Channel Partner to be able to use the data for developer billing, revenue share, and settlement reconciliation (note this is not the settlement in itself, but the data that underpins such an invoice). The difference in the usage event details provided between the two models is that the pricing section is not mandatory in the Wholesale model (could be included depending on the particular agreement between the Operator and the Channel Partner to contain the wholesale rate) but is required for the Marketplace Aggregator model. In this section we enumerate the minimum list of attributes that shall be exposed to the Channel Partner depending on the type of information to be exchanged. This information is considered as the minimum amount of information that would allow Channel Partners to create bills and expose usage to Developers and enable the reconciliation of service usage between Channel Partners and Operators. The usage exchange is at Product level between Operator and Channel Partner, where the Product Usage is built upon internal Operator's relevant Service Usage events that are not shared with the Channel Partner. As a general principle, due to privacy reasons, the information to be exposed doesn't include any final user data that could be considered as personal information (for instance MSISDN).

Exposed usage information to Channel Partners

Atomic Product Usage Attributes

Below is the minimum list of attributes of an atomic Product Usage:

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Name	TMF635	Optionality	Description	Comment
ID	id		Unique identifier of the usage event within Operator scope	
Creation Date	creationDate		Date of the creation of the productUsage, not equal to date of the API invocation or notification of the service. The date of the API invocation or notification of the service is part of the characteristics of the usage.	A completionDate attribute has been added for more precision on the productUsage date when status is final.
Channel Partner ID	relatedParty		The link to the Channel Partner.	TMF635 uses relatedParty and RelatedPartyRefOrPartyRoleRef to describe relationship. The Operator keeps track of what Channel Partner owns the individual usage event. It is implicitly related to the productUsage as part of the file name and is not duplicated in each productUsage record of the exportJob.
Application ID	relatedResource		The ID of the Application given by Operator after TMF931 POST /application	
Product ID	product		The product that the event was guided to.	TMF635 uses Product reference to describe the relationship.

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Name	TMF635	Optionality	Description	Comment
Product Usage Specification ID	usageSpecification		The usage spec that is associated with the usageCharacteristics that is used to define the characteristics of the productUsage (see below).	TMF635 uses ProductUsageSpecificationRef to describe the association. For TMF937 first version, we omit this and add it as needed in future versions of the API.
Usage Type	usageType		This attribute can be used to refine the type of usage, for instance in the case one single ProductUsageSpecification manages different types of usages.	For TMF937 first version, we omit this and add it as needed in future versions of the API.
Status	status		If this is for Wholesale, the status is notRated and if it is Marketplace Aggregator model it is rated. If rated, we also have the TMF767 UsageProductPrice attributes included in the event.	
Rating information	usageProductPrice	conditional	Prices assigned to the product usage.	Only required for Marketplace Aggregator model where a price is applied. The mandatory attribute for price is dutyFreeAmount (tax exclusive) considering that this is a B2B2X scenario. In case it is required later to have taxes in the price it can be added in a backward compatible way.

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Values of the attributes for a given API and operation are defined in characteristics that relate to the Product Usage Specification relevant to the particular CAMARA API. Some characteristics are common across all APIs and should be included in all Product Usage Specs:

- Correlation ID (x-correlator in CAMARA API header) if provided in the CAMARA API request. x-correlator is conditionally mandatory if the x-correlator is provided it has to be included and mapped with a standardized name for it to facilitate interoperability.
- Purpose - For TMF937 first version, we will omit this and add it as needed in future versions of the API.
- Scope - For TMF937 first version, we will omit this and add it as needed in future versions of the API.

Characteristics inherited from the CAMARA API invocation should retain the naming convention from the CAMARA API to facilitate interoperability and understanding. CAMARA API characteristics that are Personally Identifiable Information (PII) or delivery related are not to be included as characteristics.

Mechanism for Usage Exchange

There is a need to exchange the Usage data between Operator and Channel Partner in an efficient and timely way considering that it is a large volume of data. It needs to support various use cases such as reporting for an Application Owner, internal analytics for Channel Partner, dashboards, operational support including complaints, etc. The key requirement is to provide a mechanism to exchange a bulk of usage data for settlement reconciliation and billing. In the future, online queries and on-demand reporting may be considered.

Bulk Transfer

TMF937 adopts a bulk export approach, modeled on the TM Forum ExportJob pattern and aligned with GSMA BCE (Billing and Charging Evolution) practices. This pattern is especially important in high-volume scenarios like CAMARA API monetization.

Frequency of data exchange

Operators should be able to export usage records on a regular schedule — such as hourly, daily, or weekly — depending on partner needs and system capabilities. This flexibility allows the model to support both near-real-time usage and end-of-day settlement workflows. In the first version it is assumed that the schedule is part of the Agreement and is set up by the Operator. In future versions we may also include the possibility to set up on-demand subscriptions by the Channel Partner.

File Structure and Naming Convention

The exchanged Product Usage is compressed and include both the Product Usage data, metadata and include security through signatures.

Metadata

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Field Name	Description	Type	Length	Considerations for TMF937
Version	File template version – mandatory	Alpha-numeric	min 5	Format to use: x.y.z
ID	The ID of the exportJob that generated the file to keep the link between exportJob and file generated (traceability).	UUIDv4	36	
Status	Identifies whether the file is for testing or commercial stage. Values: C – Commercial report, T – Test report	Alphabetic	1	
Report Type	Indicates whether the file to be exchanged is supporting bilateral agreement or a dispute of previously exchanged aggregated report. Values: IN – Initial regular file, .. – TBD in future	Alphabetic	2	IN = Initial regular file; other values TBD based on future use cases.
Serving Party	Identifier of the Operator through the short ID defined in the Agreement between Operator and Channel Partner (Operator ID).	Alpha-numeric	5	The onboarding ID is allowed to be up to 200 characters. This is not practical for a filename, hence the suggestion to have a short ID.
Served Party	Identifier of the Channel Partner through the short ID defined in the Agreement between Operator and Channel Partner (Channel Partner ID).	Alpha-numeric	5	Same rationale as Serving Party.

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Field Name	Description	Type	Length	Considerations for TMF937
File Date Interval	This date range represents the time period referred by the report, not the usage contained in the report. The date used is the completionDate.	Alpha-numeric	41	Format: YYYYMMDDThhmmss.mmmZ-YYYYMMDDThhmmss.mmmZ; FROM is inclusive, TO is exclusive.
Machine readable file date interval	Machine readable – generated from the human readable date in previous row; need two fields – one start and one end.	Large integer	64 bits (8 bytes)	Equivalent to the Linux epoch (elapsed seconds since 1970-01-01T00:00:00Z) extended to 64 bits.
Earliest Reported Date and Time	Earliest Serving Network timestamp for traffic contained within the report (start date of the real usage).	Alpha-numeric	19	ISO 8601 basic format (YYYYMMDDThhmmssZ); not populated when no data found.
Latest Reported Date and Time	Latest Serving Network timestamp for traffic contained within the report (start date of the real usage).	Alpha-numeric	16	ISO 8601 basic format (YYYYMMDDThhmmssZ); not populated when no data found.
File Creation Timestamp	The date and time when file is created by the sending party.	Alpha-numeric	16	ISO 8601 basic format (YYYYMMDDThhmmssZ)
Number of Records	The total number of ProductUsage records.	Integer	Variable	If the number of records is 0 then it is a Notification file which contains only header and trailer information.

File Contents

Each exported file includes zero or more ProductUsage records. If no usage event exists for a reported time period, a file is produced but it contains 0 ProductUsage indicating an empty file. These records reference a ProductUsageSpecification, defined separately in the catalog via TMF936 (planned in a future version), so partners understand how to interpret the usage event characteristics. Records may optionally include pricing details (usageProductPrice) — required in Marketplace Aggregator models where the Operator handles first level rating.

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Date and Time Stamps

To ensure consistency and avoid ambiguity across systems and time zones, all dates and times in filenames and attributes are expressed in ISO 8601 basic format (YYYYMMDDThhmmssZ) using UTC (denoted by Z), except fileDateInterval that contains milliseconds for best accuracy (YYYYMMDDThhmmss.mmmZ). This choice eliminates issues with local time zones, daylight savings, and platform-specific filename restrictions, while providing second-level precision and a globally recognized standard for timestamp representation.

Mutual TLS (Two-Way SSL) Authentication for Report File Consumption

The system shall enforce two-way SSL (mutual TLS) authentication for all HTTPS requests made to the /exportJob endpoint. This ensures that both the data server and the consuming Channel Partner authenticate each other before any report files are exchanged and that the Channel Partner only has access to the data pertaining to their agreement with the Operator.

- During onboarding, each Channel Partner must register a public certificate with the Operator.
- When invoking /exportJob, the partner must use the corresponding private key to establish the TLS session.
- The server must validate the presented certificate against the registered public key.
- Requests from unrecognized or invalid certificates must be rejected.
- The validated partner identity must be used as a claim in report generation to guarantee the file is associated only with the authenticated partner.
- All report downloads must be served only over authenticated and encrypted connections.

For transferring sensitive files, HTTPS and SFTP can both be very secure if configured correctly, but for pure system-to-system file exchange SFTP is usually preferable because it is purpose built for file operations and naturally encourages strong key-based authentication and end-to-end encryption. HTTPS can be equally strong cryptographically and is the better choice when you need browser access or integration with existing web/API security (OAuth, mTLS, WAFs, etc.).

For now, only SFTP is considered for TMF937 first version.

Download links should always be short-lived and bound to the authenticated API client who requested them by using JWT time-limited tokens. Additionally, access to the file itself must require prior authentication so that the link cannot be used without verifying the API client identity.

File Retention and Availability

In the agreement between the Operator and the Channel Partner, the retention period of the files has to be specified. This retention period relates to the period the files are made available to the Channel Partner and not the retention period of the content based on regulation (this is up to the Operator's policy and local regulation). This is the basis for how the Operator makes the files available in a sliding window for the Channel Partner to consume the files. The retention period of the files is the same as the retention period of the exportJob. Export history is recommended, so previous exports within the pre-defined

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retention period can be re-accessed.

Notifications (can be planned for future version)

Operators should notify partners when usage data is available. Channel Partner can subscribe to ExportJob related events via Hub resource or regularly poll the exportJob resource. The notification should include download instructions or a secure link to the file, along with basic metadata (like date range and batch ID).

Reconciliation between Operator and Channel Partner

For both Wholesale Reseller and Marketplace Aggregator models, sufficient details need to be provided to be able to reconcile the use of the service reported as Product Usage with the amount owed based on the API Usage and product definition. For Wholesale Reseller agreements the Operator invoices the Channel Partner based on the API Usage for a given period at the agreed wholesale rate. For Aggregated Marketplace agreements, the Channel Partner provides a statement of the calculation of the revenue share that is to be deposited to the Operator.

Use Cases

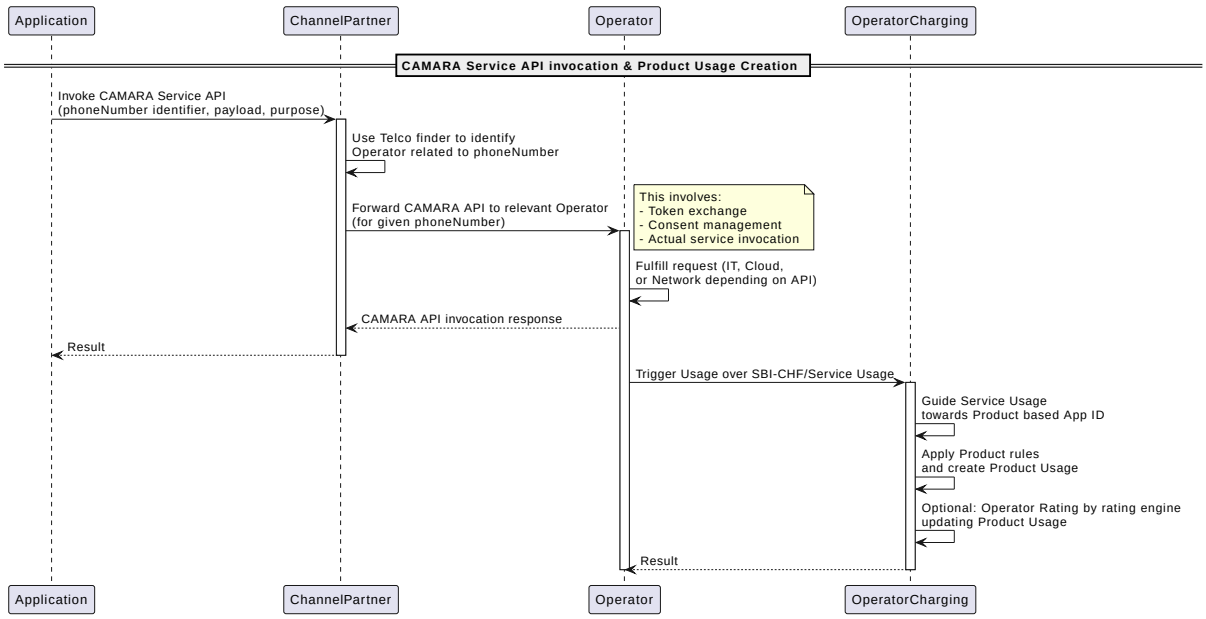
All use cases apply on Wholesale Reseller and Aggregated models - to any Channel Partner.

Creation of Product Usage (Pre-requisite)

This use case describes the E2E flow of how usage is created to enable the exposure of the usage data to the Channel Partner. It is in the context of TMF937 a pre-requisite to explain the data creation prior to exchanging the data from Operator to Channel Partner.

- Developer invokes a CAMARA API operation via Channel Partner.
- Channel Partner forwards request to Operator.
- Operator invokes the actual service and receives enriched response.
- Operator forwards the response to the application and sends enriched charging request internally.
- Operator charging guides the API to the appropriate product and applies product rules according to the product catalog definition including the definition of the characteristics of the event based on the associated Product Usage Specification.
- A ProductUsage record is created, including:
  - Party identifiers (Application, Channel Partner)
  - CorrelationID
  - API event characteristics
  - Pricing information (if applicable)

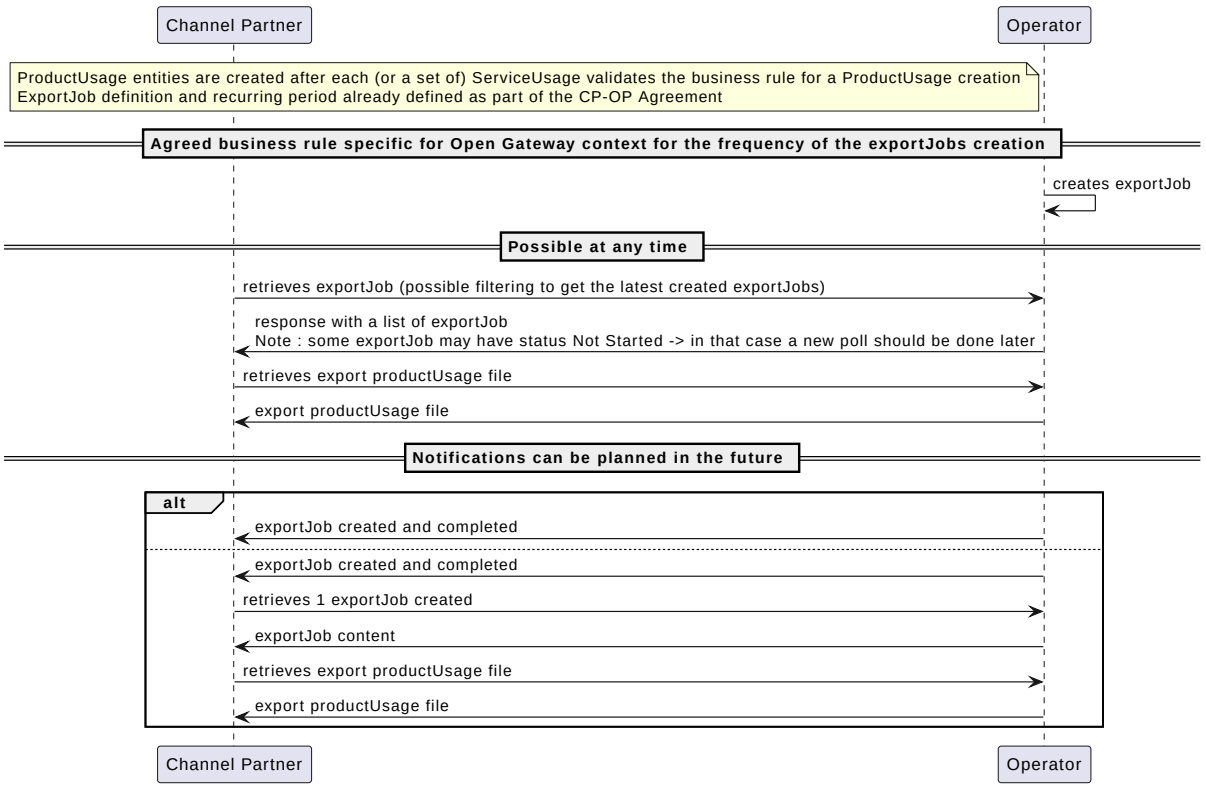
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Creation of file that contains a list of Product Usage and retrieval of the file by Channel Partner

Assumption: the retrieval of the bulk is performed using SFTP. Notifications can be planned in a future version.

- Operator creates a file that contains a list of ProductUsage created during a time window (export of the ProductUsage resource).
- Channel Partner retrieves ProductUsage export file.



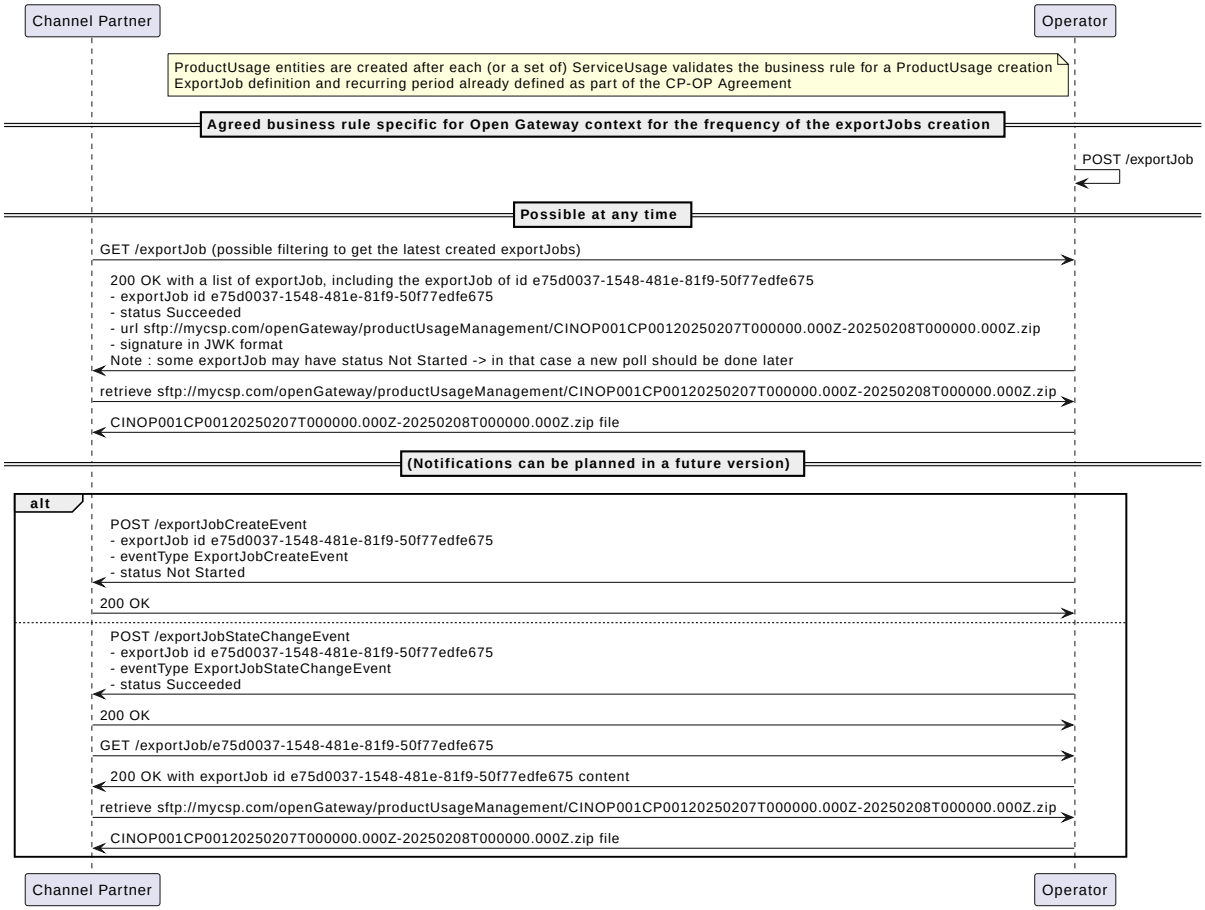
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Sequence diagrams

Creation of file that contains a list of Product Usage and retrieval of the file by Channel Partner

Assumptions:

- Bulk file retrieval is performed over SFTP.
- Partner authentication uses mutual TLS.
- (Can be planned for future version) Notifications are delivered through REST APIs (for example, POST ExportJobCreateEvent).
- The exportJob query string is done using productUsage.completionDate as the reference date for the productUsage. completionDate is the date when the productUsage reached a final status (rated or notRated).



ExportJob payload example

```
{
  "id": "8e8ce309-835a-4f4d-8ca2-0b33467e8fac",
  "href": "https://mycsp.com/tmf-api/openGatewayProductUsage/v5/exportJob/8e8ce309-
835a-4f4d-8ca2-0b33467e8fac",
  "@type": "ExportJob",
  "path": "/productUsage",
  "query": "status=rated,notRated&completionDate.gte=2025-02-
07T00:00:00.000Z&completionDate.lt=2025-02-08T00:00:00.000Z",
```

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```
  "creationDate": "2025-02-08T00:01:00Z",
  "completionDate": "2025-02-08T00:02:35Z",
  "contentType": "application/gzip",
  "url":
"sftp://mycsp.com/openGateway/openGatewayProductUsage/CINOP001CP00120250207T000000.0
00Z-20250208T000000.000Z.zip",
  "signature":
"eyJhbGciOiJSUzI1NiIsDQoieDVjIjogWyANCiJNSUlCSWpBTkJna3Foa2lHOXcwQkFRRUZBQU9DQVE4QU1
JSUJDZ0tDQVFFQTF0UHZ3eWRQUzJOMVI4VVcyM1hzQzJ3eXdqSEdTUUZlbGVkTmNhSHV5aF95NHB6SjhvbDd
OZHp2bzRQNmF6dGNjRnZDY3VXekd0WHY3TmhZazJKd24xWWtmdnVBTS1qUTV6MjlYS2F6YThZcEtxUk5USTB
vRWZSRndVQ3BtVF9KYk5kMnQydUdXMzNGVWI3MHI1Z3AwRmw1cUNOX2g1cU03Y1g2RDVUM0hWeXRXbFc0ZlV
VTkhpVG9nZnJHanNjS3pDb0ZOdmtFeFRTUTdocW5ReGdzY1BBZ0Y4VGFmNkd2TzhBQUFxa2I1X25JYmNxZFU
xQlZVSFMyVVdISHVNR2JLMkxUVXE5bEFybGZDY2Q5NlItVWtNQXBGbFZQLTRraTJReWROanFhU2FrU1ZWbmt
YU3M5bmZQVnlMZWdyd2Yxcl8tZ19QdzlDQ2tjUGgyT3JXM2g2d0lEQVFBQiINCl0NCn0..B_YtCrSnlsEiqR
AcwO1-S0YHp486I8I4te9D-QzF2y0nKrt-
v6l9pzp9TngVrbkJ42shoUyiCsT5bet2HsQLJiBT1bQWOlf3unVEx17Z2sTGr1tZml4M2KWUoNvdVuqyGB8c
o43taSGnQ05zeRfs-C7JD_Z_cx6byhCbyhr-
bmTiF1St7lUYbxVDwpCMl7gAf4TQdM48inmQGRKjjEBHXJKhaBXcADaTiyDlEc1K8eWNGoxhxaJWJQx0yDkb
p2YhxNKBXhWDTBhvsRIIwCSFFFT9LjiXo8ouaSrDA-
2wrZz2NEarl4cc4nZTUL8ducMCJYDUmFnHVF6jb03G4MrVjQ",
  "status": "Succeeded"
}
```


File Packaging, Naming, and Metadata

Defines the physical structure, naming conventions, and metadata content for the ProductUsage bulk exchange files exchanged through /exportJob.

File Packaging Structure

Each export job produces a ZIP archive containing:

Component	Description
JSON	Zero or more JSON files with ProductUsage records. It is the equivalent of the result of a GET /productUsage using the query string passed in the exportJob `query` attribute.
metadata	JSON file with metadata describing the batch (see schema below).

Zip file with its content :

```
.zip
 ¬¬ *.json
 ¬¬ *.metadata
```

The ZIP file contains JSON file(s) and metadata file. The JWS (JSON Web Signature) signature is embedded as a field in the ExportJob (see example above).

Exception Cases

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Case	Behavior
Missing ZIP for expected interval	Indicates a missing file in the chronological sequence; requires reconciliation. The rules to define the intervals and contiguity for each ExportJob are based on an agreement between Channel Partner and Operator. It is the Operator that puts the values in the POST /exportJob for each ExportJob, so is responsible if there is a missing file.
No ProductUsage records produced	ZIP file still created; metadata reports `"numberOfRecords": 0`.
JSON too large	Operator and Channel Partner agree on maximum JSON size or record count; split into multiple JSON files (`_00001.json`, `_00002.json`, etc.).
ZIP too large	Operators may adjust the time interval or compression level as a mitigation to avoid too large ZIP file. (Future enhancement note in requirements.)

File Naming Convention

The `url` and `signature` attributes must reference files whose names comply with the TMF937 file-naming standard defined for ProductUsage ZIP exports.

For zip, metadata and signature:

`<Status><ReportType><ServingParty><ServedParty><FileDateInterval>[<_SequenceNumber>].<extension>`

Element	Meaning
Status	C = Commercial, T = Test
ReportType	IN (initial); future types TBD
ServingParty	5-character Operator ID
ServedParty	5-character Channel Partner ID
FileDateInterval	UTC start–end timestamps (ISO 8601 with milliseconds)
_SequenceNumber	Only applicable for JSON files.

Note: The time period (FileDateInterval) is based on the ExportJob query and is also the attribute used to indicate a missed period. For each time period there should be a ZIP file. So the Channel Partner must be able to check if there are missing file date intervals based on the FileDateInterval attribute of all files. There should be no gaps.

Examples:

Single-file (vanilla) case:

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```
CINOP001CP00120250207T090000.000Z-20250208T000000.000Z.zip
  CINOP001CP00120250207T090000.000Z-20250208T000000.000Z_00001.json
  CINOP001CP00120250207T090000.000Z-20250208T000000.000Z.metadata
  CINOP001CP00120250207T090000.000Z-20250208T000000.000Z.sign
```

Multiple JSON files:

```
CINOP001CP00120250207T090000.000Z-20250208T000000.000Z.zip
  CINOP001CP00120250207T090000.000Z-20250208T000000.000Z_00001.json
  CINOP001CP00120250207T090000.000Z-20250208T000000.000Z_00002.json
  CINOP001CP00120250207T090000.000Z-20250208T000000.000Z.metadata
  CINOP001CP00120250207T090000.000Z-20250208T000000.000Z.sign
```

Zero-record case:

```
CINOP001CP00120250207T090000.000Z-20250208T000000.000Z.zip
  CINOP001CP00120250207T090000.000Z-20250208T000000.000Z.metadata
  CINOP001CP00120250207T090000.000Z-20250208T000000.000Z.sign
```

Metadata Schema (MVP)

This schema is hosted under GSMA schema repository, see draft publication here: https://github.com/GSMA-Open-Gateway/Operate-API-References/blob/ProductUsage-v5.0.0/Schemas/Gsma/ProductUsageExportFileMetadata/ProductUsageExportFileMetadata.schema.json

```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "$id": "ProductUsageExportFileMetadata.schema.json",
  "title": "TMF937 ProductUsage export file Metadata Schema",
  "description": "File-level metadata for exchange of CAMARA API Product Usage
between a Serving Party (Operator) and a Served Party (Channel Partner).",
  "type": "object",
  "properties": {
    "metadata": {
      "type": "object",
      "required": [
        "version",
        "id",
        "status",
        "reportType",
        "servingParty",
        "servedParty",
        "fileDateInterval",
        "machineReadableFileDataInterval",
        "earliestReportedDateTime",
        "latestReportedDateTime",
        "fileCreationTimestamp",
        "numberOfRecords"
      ],
      "properties": {
        "version": {
          "type": "string",
```

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```
          "pattern": "^[0-9]+\\.[0-9]+\\.[0-9]+$",
          "description": "Semantic version of the metadata schema (e.g., 1.0.0).",
          "examples": [ "0.0.1", "1.0.0", "10.1.2" ]
        },
        "id": {
          "type": "string",
          "format": "uuid",
          "description": "Unique identifier for this report file instance (UUID
format). It is same value as the exportJob.id that generated the file; this permits
to keep the link between export job and file generated (traceability).",
          "examples": [ "8e8ce309-835a-4f4d-8ca2-0b33467e8fac" ]
        },
        "status": {
          "type": "string",
          "enum": [ "C", "T" ],
          "description": "Report status: C = Commercial, T = Test."
        },
        "reportType": {
          "type": "string",
          "enum": [ "IN" ],
          "description": "Report type identifier (e.g., IN for initial). Future
values to be determined"
        },
        "servingParty": {
          "type": "string",
          "pattern": "^[A-Z0-9]{5}$",
          "description": "5-character code identifying the serving operator.",
          "examples": [ "OP001" ]
        },
        "servedParty": {
          "type": "string",
          "pattern": "^[A-Z0-9]{5}$",
          "description": "5-character code identifying the served channel partner.",
          "examples": [ "CP001" ]
        },
        "fileDateInterval": {
          "type": "object",
          "required": [ "startDateTime", "endDateTime" ],
          "properties": {
            "startDateTime": {
              "type": "string",
              "format": "date-time",
              "description": "UTC start timestamp of the reporting interval (ISO
8601) including milliseconds.",
              "examples": [ "2025-02-07T00:00:00.000Z" ]
            },
            "endDateTime": {
              "type": "string",
              "format": "date-time",
              "description": "UTC end timestamp of the reporting interval (ISO 8601)
including milliseconds.",
              "examples": [ "2025-02-08T00:00:00.000Z" ]
            }
          },
          "description": "The reporting period covered by the file, expressed in
UTC."
        },
        "machineReadableFileDataInterval": {
          "type": "object",
          "required": [ "startEpoch", "endEpoch" ],
          "properties": {
            "startEpoch": {
              "type": "integer",
```

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```
              "description": "Unix epoch time (seconds) corresponding to
fileDateInterval.startDateTime.",
              "examples": [ 1738780800 ]
            },
            "endEpoch": {
              "type": "integer",
              "description": "Unix epoch time (seconds) corresponding to
fileDateInterval.endDateTime.",
              "examples": [ 1738876800 ]
            },
            "note": {
              "type": "string",
              "description": "Optional comment; if ISO 8601 is universally used,
this section may be deprecated."
            }
          },
          "description": "Machine-readable equivalent of fileDateInterval; typically
epoch timestamps."
        },
        "earliestReportedDateTime": {
          "type": "string",
          "format": "date-time",
          "description": "Earliest actual serviceInvocationDateTime reported in the
file (UTC). It corresponds to the start date of the real usage (service usage).",
          "examples": [ "2025-02-06T23:59:58.325Z" ]
        },
        "latestReportedDateTime": {
          "type": "string",
          "format": "date-time",
          "description": "Latest actual serviceInvocationDateTime reported in the
file (UTC). It corresponds to the start date of the real usage (service usage).",
          "examples": [ "2025-02-07T23:56:51.675Z" ]
        },
        "fileCreationTimestamp": {
          "type": "string",
          "format": "date-time",
          "description": "Timestamp of file creation; must be later than
fileDateInterval.endDateTime.",
          "examples": [ "2025-02-08T00:02:30Z" ]
        },
        "numberOfRecords": {
          "type": "integer",
          "minimum": 0,
          "description": "Total count of data records contained in the file.",
          "examples": [ 738264 ]
        }
      }
    }
  },
  "required": [ "metadata" ]
}
```

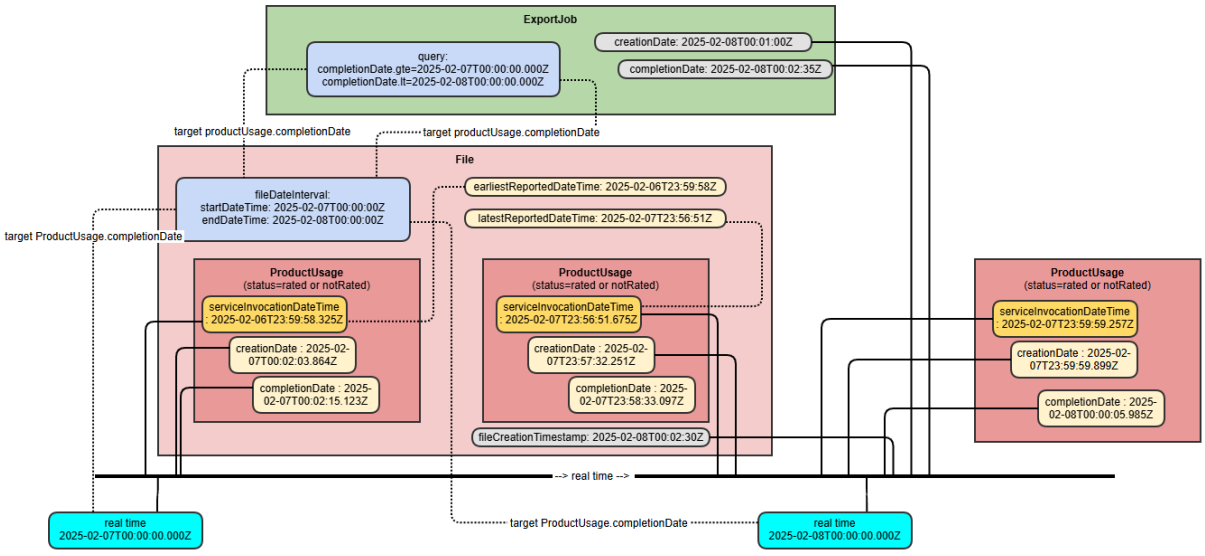
Example of output:

```
{
  "metadata": {
    "version": "1.0.0",
    "id": "2f9a18b1-8c5a-4a2b-9e4b-9a6e2f7f6a1d",
    "status": "C",
    "reportType": "IN",
```

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```
    "servingParty": "OP123",
    "servedParty": "CH987",
    "fileDateInterval": {
      "startDateTime": "2025-02-07T00:00:00.000Z",
      "endDateTime": "2025-02-08T00:00:00.000Z"
    },
    "machineReadableFileDataInterval": {
      "startEpoch": 1738886400,
      "endEpoch": 1738972799,
      "note": "Unix epoch (seconds since 1970-01-01T00:00:00Z)"
    },
    "earliestReportedDateTime": "2025-02-06T23:59:58Z",
    "latestReportedDateTime": "2025-02-07T23:56:51Z",
    "fileCreationTimestamp": "2025-02-08T00:02:30Z",
    "numberOfRecords": 125436
  }
}
```


Examples on a diagram



Alignment with Metadata

The metadata file must reflect and confirm all attributes encoded in the filename and referenced by the ExportJob.

Alignment Element	Source	Validation Rule
servingParty	Metadata JSON	Must match ServingParty in filename
servedParty	Metadata JSON	Must match ServedParty in filename
fileDateInterval.startDateTime / endDateTime	Metadata JSON	Must match FileDateInterval in filename
status	Metadata JSON	Must equal filename Status prefix
reportType	Metadata JSON	Must equal filename ReportType

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If mismatch occurs, the ExportJob transitions to `status = Failed`, and an error is reported.

File content

Product Usage

In the context of **TMF937**, a **ProductUsage** record represents the structured report of a service that has been delivered, guided to a product and is considered billable. It captures the essential information needed for **monetization, settlement, and reconciliation** between an Operator and a Channel Partner.

Each ProductUsage record is defined by a corresponding **ProductUsageSpecification**, which standardizes **what type of usage occurred** (e.g., API call, session, data query) and its associated **characteristics** (e.g., correlation ID, duration, geo scope). This separation ensures consistent modeling across different offers and CSP implementations.

While **ProductUsage is generally agnostic of pricing logic**, its role **depends on the monetization model**:

- In a **marketplace aggregator model**, the operator performs the rating, and the ProductUsage record will **include priced values** (e.g., cost, tier applied, currency) to reflect the final charge to the channel partner or developer.
- In a **wholesale model**, the ProductUsage may be exchanged as a processed **usage record** with the relevant characteristics, but with pricing determined by the aggregator.

TMF937 ensures that this usage data can be **exchanged efficiently, transparently, and interoperably**, supporting:

- Standardized usage structures across parties
- Optional inclusion of pricing context when required
- Accurate and auditable monetization across B2B2X ecosystems

In TMF937, ProductUsage is not created or modified by the API — it is *exported*. The Operator's systems remain the system of record; TMF937 defines only the **format and delivery mechanism** for exchanging usage data with Channel Partners. This model enables CSPs to scale participation in Open Gateway by aligning **technical API behavior with commercial execution**, and providing the data foundations for **automated rating, invoicing, and dispute resolution**.

TM Forum References:

- SID ABE: *Product Usage*
- Component : TMFC040 ProductUsage component
- TMF635 v5.0.0 draft — Product Usage Management API (for ProductUsage)
- TMF767 v5.0.0 draft — Product Usage Catalog Management API (for ProductUsageSpecification)
- TMF620 v5.0.0 — Product Catalog Management (for ExportJob)

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Resource

The resource that is the basis of the TMF937 API is ProductUsage as defined in TMF635 v5.0.

TMF937 is responsible for exchanging the data between the Operator and the Channel Partner. How the ProductUsage is created and maintained is internal to the Operator and is not applicable for TMF937. What usageProperties that is applicable for a given usage event is determined by the ProductUsageSpecification as part of the ProductOffering.

UsageProductPrice is only required for Marketplace Aggregator model cases where prices are determined by Operator.

Excluded attributes from base schema

Attribute	Exclusion Rationale
href	no GET /productUsage operation yet so would not be valued
usageType	no requirement
usageCharacteristic	replaced by usageProperties for strongly typed characteristics
relatedParty	no requirement
externalIdentifier	no requirement
usageSpecification	no need for MVP as the specification is materialized in the schema referred via schemaLocation. May be added in the future if there are requirements.
serviceUsage	Internal reference to network/service layer usage; not relevant to Channel Partner.
usageProductPrice.priceAlteration usageProductPrice.productOfferingPrice usageProductPrice.recurringChargePeriod usageProductPrice.unitOfMeasure usageProductPrice.priceType usageProductPrice.name	Simplified out of MVP; price per record captured in UsageProductPrice if applicable. No need for additional information.
usageProductPrice.price.percentage usageProductPrice.price.taxIncludedAmount usageProductPrice.price.taxRate	Not relevant in B2B2X exchange; all pricing duty-free and tax applied at later billing stages.

Future versions may consider reintroducing some of these attributes as optional, depending on Open Gateway marketplace models.

Additional attributes not in base schema

Attribute	Exclusion Rationale
completionDate	date when the productUsage has reached a final status (rated or notRated)

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Attribute	Exclusion Rationale
application	reference to the Application, using ApplicationRef schema
usageProperties	replaces usageCharacteristics for stronglyTyped definition of characteristics

Structure and Relationship

Each `ProductUsage` record references a ProductUsageSpecification that defines the expected characteristics, such as usage type, reporting dimensions, and correlation keys. TMF937 packages multiple ProductUsage records into a bulk exchange file, accompanied by metadata describing the batch context (period, count, and serving/served parties).

```
ProductUsage 1..* ¬ definedBy ¬ ProductUsageSpecification 1
ProductUsage 1..1 ¬ exportedThrough ¬ ExportJob 1
```

Lifecycle and Exchange Status Policy

This section defines the valid **status values** for ProductUsage and clarifies **which states are eligible for exchange** between the Operator (Serving Party) and Channel Partner (Served Party). The intent is to ensure that only finalized and commercially meaningful usage data is shared, while intermediate or error states remain internal to the Operator.

The standard ProductUsage lifecycle is defined in TMF635, including operational and validation states.

The Product Usage that is to be shared between the Operator and the Channel Partner should be as follows:

Model	Description	Shared Status
Marketplace Aggregator Model	Operator performs rating before exchange; shared usage includes monetary values.	rated
Wholesale Model	Operator shares validated, unrated usage; aggregator/partner applies own pricing.	notRated

Any other status is of internal interest of the Operator and should not be exchanged with the Channel Partner. The following shows all statuses with comments on relevance:

Status	Meaning	Exchange Visibility	Comments
initialized	Usage record created, validation pending	□ Internal	Not complete; not shared
rejected	Validation failed	□ Internal	Excluded from exchange
validated	Format and business rules passed; ready for rating	□ Internal	Pre-rating only

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Status	Meaning	Exchange Visibility	Comments
corrected	Correction applied to a previously failed record	□ Internal	May later progress to rated/notRated
rated	Successfully rated; final price known	□ External (Marketplace model)	Included in exchange when Operator determines price
notRated	Validated but not rated; price determined by partner	□ External (Wholesale model)	Included in exchange when pricing handled externally
failed	Rating failed	□ Internal	Operator-only
superseded	Usage replaced by a newer record	□ Internal	Do not exchange obsolete data

The filtering of eligible ProductUsage records by status is defined in the `ExportJob.query` parameter of the TMF937 `/exportJob` operation. Example: `query: status=notRated,rated&completionDate.gte=2025-08-10T00:00:00.000Z&completionDate.lt=2025-09-09T59:59:59.999Z`

In TMF937, only those records with `status=rated` or `status=notRated` are included in the exported file.

Polymorphism

In the Open Gateway context, the ProductUsage are modelled using strongly-typed schemas, see more details in ProductUsageSpecification chapter.

ProductUsage Example

This ProductUsage example corresponds to a CAMARA Number Verification v1 – POST /verify. Note that this is a proposal based on the latest draft schema; the final attributes will be based on the final standardized schema that will be available in the GSMA repository.

Wholesale Model (notRated)


```
{
  "id": "79e8c5b5-85da-4e35-9102-851bf66a15ae",
  "@type": "OpenGatewayNumberVerificationVerifyProductUsage-v1.0.0",
  "@baseType": "ProductUsage",
  "@schemaLocation": "https://mycsp.com/tmf-
api/schemas/v5/OpenGatewayNumberVerificationVerifyProductUsage-v1.0.0.schema.json",
  "status": "notRated",
  "creationDate": "2025-02-07T00:02:03.864Z",
  "completionDate": "2025-02-07T00:02:15.123Z",
  "usageProperties": {
    "serviceInvocationDateTime": "2025-02-06T23:59:58.325Z",
    "x-correlator": "123e4567-e89b-12d3-a456-426614174000"
  },
  "product": {
    "@type": "ProductRef",
    "@referredType": "ApiProduct",
    "id": "8bb8bcdb-9fcf-4b4f-8b27-2dda848404fb"
```


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```
  },
  "application": {
    "@type": "ApplicationRef",
    "@referredType": "Application",
    "id": "f1cba17d-789a-4037-b3e3-2c96e887576c"
  }
}
```

Marketplace Aggregator Model (rated)

(Same usage event, identical characteristics)

```
{
  ...
  "usageProductPrice": [
    {
      "@type": "OpenGatewayProductPrice",
      "price": {
        "@type": "OpenGatewayPrice",
        "dutyFreeAmount": {
          "value": 0.01,
          "unit": "EUR"
        }
      }
    }
  ],
  ...
}
```

ProductUsageSpecification (not part of the OAS)

NOTE For Open Gateway TMF937 MVP, it has been decided

- to not require exposure of ProductUsageSpecification. The productUsage specification is materialized by the strongly-typed schemas of ProductUsage using polymorphism.
- all Operators have to use standard schemas available in GSMA-Open-Gateway/Operate-API-References: Repository to store schemas and information from the TMF Operate APIs. To be read in conjunction with TMF Operate APIs for Camara API usages. So that it permits all Operators to use exactly same attribute names for the usage characteristics of each Camara API.
- the definitions of these schemas are done under Operate API stream within TM Forum in collaboration with GSMA
- each Operator can decide to extend the standard schemas with additional attributes if needed, but the standard required ones must be included and not modified (same name and same semantics) to ensure interoperability across Operators and Channel Partners.

Definition

A **ProductUsageSpecification** defines the reusable structure, semantics, and measurement dimensions of a ProductUsage record. It acts as the **blueprint** for how a specific type of usage event (e.g., API call, session, query) is captured, identified, and interpreted for charging, analytics, and settlement purposes.

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Each ProductUsageSpecification is:

- **Reusable** across multiple ProductOfferings, even when pricing or allowances differ.
- **Independent** of commercial terms (e.g., tiers, bundles, discounts).
- **Stable** across partners and markets, enabling interoperability and catalog modularity.

In TMF937, ProductUsageSpecification defines the **structure of the exchanged usage records**, ensuring that both Operator and Channel Partner interpret the usage data consistently. It is not exchanged dynamically via the API but serves as a **reference definition** to interpret ProductUsage attributes within each file.

TMF References

- **TMF767** – Product Usage Catalog Management API
- **TMF620** – Product Catalog Management API
- **SID ABEs:** Product Usage Specification, Usage Specification Characteristics

Business Example

A single `geo.session.v1` ProductUsageSpecification for a Geofencing API can support multiple offerings:

Offering	Model	Description
Basic Geofence	Pay-as-you-go	Charged per session/minute
Pro Geofence	Tiered bundle	Monthly allowance + overage
Enterprise Geofence	SLA bundle	Fixed-price tier with included usage

All share the same usage specification (structure, attributes, units), but differ in pricing logic defined elsewhere in the catalog.

Resource (can be planned for future release)

See TMF767 v5.0.0 — Product Usage Catalog Management API for details on the ProductUsageSpecification resource.

Excluded attributes from base schema (can be planned for future release)

Attribute	Reason
serviceUsageSpecification	Internal to Operator; not relevant to Channel Partner.
attachment	Not applicable; no external specification documents required in MVP.
specCharacteristic	Not applicable for strongly typed

Excluded added to base schema (can be planned for future release)

No need for extension.

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Product Usage Specification Characteristics

For MVP, TMF937 defines usage specifications for Number Verification v1 and SIM Swap v1 (aligned with CAMARA APIs).

No PII or delivery attributes are included as characteristics.

Each row below represents one *API operation* mapped to the CAMARA API and its invocation, with non-PII and delivery characteristics included. The reference of the standard characterictis chosen in Open Gateway is the GSMA github repository GSMA-Open-Gateway/Operate-API-References: Repository to store schemas and information from the TMF Operate APIs. To be read in conjunction with TMF Operate APIs. Below you find examples for illustration:

API	ProductUsageSpecification name	Type of Event	Characteristics *(CAMARA attributes minus PII/Delivery)*	Derived From	Notes
Number Verification v1	OpenGatewayNumberVerificationVerifyProductUsage-v1.0.0	Atomic	serviceInvocationDateTime, x-correlator, httpStatus (not for now, could be added later if required)	POST /verify	Includes xCorrelator; timestamp in ISO 8601 UTC
Number Verification v1	OpenGatewayNumberVerificationDevicePhoneNumberProductUsage-v1.0.0	Atomic	serviceInvocationDateTime, x-correlator, httpStatus (not for now, could be added later if required)	GET /device-phone-number	Same pattern; used for device identity sharing
SIM Swap v1	OpenGatewaySIMSwapRetrieveDateProductUsage-v1.0.0	Atomic	serviceInvocationDateTime, x-correlator monitoredPeriod (to be confirmed), httpStatus (not for now, could be added later if required)	POST /retrieve-date	Excludes latestSimChange (non-chargeable)
SIM Swap v1	OpenGatewaySIMSwapCheckProductUsage-v1.0.0	Atomic	serviceInvocationDateTime, x-correlator, httpStatus, httpStatus (not for now, could be added later if required)	POST /check	Includes xCorrelator; excludes PII, delivery attributes, maxAge and swapped (the last two are non-chargeable)

NOTE:

- Characteristics derived from API **response payloads** (e.g., swapped, latestSimChange, device, phonenumber, sink) are typically *not chargeable* or may be perceived as sensitive PII data or delivery attributes. These types of attributes are omitted from the Product Usage Specfication.
- The characteristics included are request-level characteristics that identify the **invocation** (time, correlation, subscription ID, chargeable attributes) and are retained for TMF937 exchange.

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- For Atomic events the HTTP Response (status) may be optionally included as a characteristic where there is a need to be to distinguish and charge for different outcomes. The norm is that all 200 responses are charged, but there might be cases such as where a request is made but the response cannot be given as a device is turned off (422) where the operator may wish to still see this as chargeable.

NOTE

Usage type is a classification of usage based on the characteristics, used for reporting, invoice grouping and more. Type of Event defines how a usage event is valued. It reflects the physical or logical dimension that the product uses to quantify consumption. For MVP we cover Atomic APIs but in the future we will cover also subscription and notification based APIs that may have other requirements.

ProductUsageSpecification example (can be planned for future release)

Note this example is based on strongly-typed pattern for ProductUsage schemas.

ProductUsageSpec Example

```
{
  "id": "e700bb98-cb37-48ed-a551-aa62cf597acb",
  "@type": "ProductUsageSpecification",
  "name": "Number Verification - verify",
  "version": "1.0.0",
  "description": "Defines structure and characteristics of ProductUsage verify for
the CAMARA Number Verification API (v1).",
  "lifecycleStatus": "active",
  "validFor": {
    "startDateTime": "2025-01-01T00:00:00Z"
  },
  "targetProductUsage": {
    "@type": "OpenGatewayNumberVerificationVerifyProductUsage-v1.0.0",
    "@schemaLocation": "https://mycsp.com/tmf-
api/schemas/v5/OpenGatewayNumberVerificationVerifyProductUsage-v1.0.0.schema.json"
  },
  "lastUpdate": "2025-02-07T00:00:00Z"
}
```

Note:

- productUsageSpecification are not really useful for the MVP as the target schema is enough to validate the payload. The @type value of the ProductUsage corresponds to a productUsageSpecification.
- target schemas are stored online in the GSMA repository GSMA-Open-Gateway/Operate-API-References
  - The draft proposals are here Operate-API-References/Schemas/Product at ProductUsage-v5.0.0 · GSMA-Open-Gateway/Operate-API-References

OpenGatewayProductUsage DRAFT common schema

```
{
  "$schema": "http://json-schema.org/draft-07/schema#",
  "$id": "OpenGatewayProductUsage.schema.json",
```

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```
  "title": "OpenGatewayProductUsage",
  "definitions": {
    "OpenGatewayProductUsage": {
      "$id": "#OpenGatewayProductUsage",
      "description": "An occurrence of usage on a product, typically derived from
one or various service usages. Specialization of TM Forum ProductUsage schema for
Open Gateway.",
      "type": "object",
      "properties": {
        "id": {
          "type": "string",
          "description": "unique identifier",
          "maxLength": 200
        },
        "@type": {
          "type": "string",
          "description": "Name of the sub class of ProductUsage in the Open Gateway
Context.",
          "maxLength": 200
        },
        "@baseType": {
          "type": "string",
          "description": "Name of the parent class. Must be ProductUsage.",
          "enum": [
            "ProductUsage"
          ]
        },
        "@schemaLocation": {
          "type": "string",
          "description": "A URI to the JSON-Schema file that defines additional
attributes, in particular the usageProperties specific to the sub-class.",
          "maxLength": 200
        },
        "creationDate": {
          "type": "string",
          "format": "date-time",
          "description": "Creation date of the ProductUsage. May be different from
the date the usage was invoked.",
          "maxLength": 200
        },
        "completionDate": {
          "type": "string",
          "format": "date-time",
          "description": "Date when the ProductUsage has reached a final status
(status=rated or status=notRated in this Open Gateway context). May be different
from the creationDate.",
          "maxLength": 200
        },
        "status": {
          "description": "Possible values for the status of the product usage. Refer
to User Guide for comprehensive life cycle diagram and descriptions of statuses.",
          "enum": [
            "rated",
            "notRated"
          ]
        },
        "usageProductPrice": {
          "description": "Price assigned to the product usage when product usage is
rated.",
          "type": "array",
          "items": {
            "$ref":
"./OpenGatewayProductPrice.schema.json#/definitions/OpenGatewayProductPrice"
```

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```
          },
          "minItems": 1
        },
        "product": {
          "$ref": "./ProductRef.schema.json#/definitions/ProductRef"
        },
        "application": {
          "description": "Identifier of the Application that is at the origin of the
product usage. This Application is referenced by the id given to the Application
during onboarding process (TMF931 POST /application)",
          "$ref":
"../Resource/ApplicationRef.schema.json#/definitions/ApplicationRef"
        }
      },
      "required": [
        "id", "@type", "@baseType", "@schemaLocation", "creationDate",
"completionDate", "status", "product", "application"
      ]
    }
  }
}
```

OpenGatewayNumberVerificationVerifyProductUsage DRAFT target schema Example

```
{
  "$schema": "http://json-schema.org/draft-07/schema#",
  "$id": "OpenGatewayNumberVerificationVerifyProductUsage-v1.0.0.schema.json",
  "title": "OpenGatewayNumberVerificationVerifyProductUsage-v1.0.0",
  "definitions": {
    "OpenGatewayNumberVerificationVerifyProductUsage-v1.0.0": {
      "$id": "#OpenGatewayNumberVerificationVerifyProductUsage-v1.0.0",
      "description": "Specific ProductUsage for CAMARA API Number Verification for
usage verify. Corresponds to API call POST /verify. @type must be valued to
OpenGatewayNumberVerificationVerifyProductUsage-v1.0.0",
      "type": "object",
      "properties": {
        "usageProperties": {
          "description": "Contains common properties for all Open Gateway Product
Usage defined as an API call",
          "type": "object",
          "properties": {
            "serviceInvocationDateTime": {
              "type": "string",
              "format": "date-time",
              "description": "The date and time when the service usage service was
invoked."
            },
            "x-correlator": {
              "type": "string",
              "description": "Service correlator to make E2E observability. Same
value as the X-Correlator in the Camara API request.",
              "pattern": "^[a-zA-Z0-9-_:;./<>{}]{0,256}$"
            }
          },
          "required": [
            "serviceInvocationDateTime"
          ]
        }
      },
      "allOf": [
        {
```

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```
          "$ref": "./OpenGatewayProductUsage-
v1.0.0.schema.json#/definitions/OpenGatewayProductUsage-v1.0.0"
        }
      ],
      "required": [
        "usageProperties"
      ]
    }
  }
}
```

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SUPPORT OF POLYMORPHISM AND EXTENSION PATTERNS

Support of polymorphic collections and types and schema based extension is provided by means of a list of generic meta-attributes that we describe below. Polymorphism in collections occurs when entities inherit from base entities, for instance an OpenGatewayExportJob inheriting properties from the ExportJob entity.

Generic support of polymorphism and pattern extensions is described in the TMF API Guidelines, Part 2 (TMF630).

The @type attribute provides a way to represent the actual class type of an entity. For example, within a list of ExportJob instances some may be instances of OpenGatewayExportJob. The @type gives this information. All resources and sub-resources of this API have a @type attributes that can be provided when this is useful.

The @referredType can be used within reference entities (like for instance a ExportJobRef object) to explicitly denote the actual entity type of the referred class. Notice that in reference entities the @type, when used, denotes the class type of the reference itself, such as OpenGatewayExportJobRef, and not the class type of the referred object. However since reference classes are rarely sub-classed, @type is generally not useful in reference objects.

The @schemaLocation property can be used in resources to allow specifying user-defined properties of an Entity or to specify the expected characteristics of an entity.

The @baseType attribute gives a way to provide explicitly the base of class of a given resource that has been extended.

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RESOURCE MODEL

The Product Usage Operate API works in the Product Usage domain of the TM Forum SID, with ProductUsage and ProductUsageSpecification as the core entities and ExportJob as the mechanism to package and deliver usage externally. File-level metadata and signatures complement these to ensure integrity, traceability and secure delivery between Operator and Channel Partner.

At this time there are no business requirements for exposing individual ProductUsage records via this API; the exposure of ProductUsage in an aggregated file is enough for now and permits to be more efficient due to high volume. There are no business requirements for exposing capability to request an exportJob (the POST exportJob operation is expected to be performed by the Operator's internal system).

Entity	Role
ExportJob	Asynchronous task that selects, packages and exposes ProductUsage data as downloadable files.

ExportJob

Definition

An ExportJob represents an asynchronous task that generates and packages finalized ProductUsage records for delivery from a Serving Party (Operator) to a Served Party (Channel Partner).

In TMF937, the ProductUsage Export capability builds upon the TMF635 Product Usage Management API v5.0, which now includes the ExportJob resource as a native, embedded capability, allowing creation, monitoring, and event publication to occur under a single API endpoint (/exportJob).

TMF937 adopts this model and specializes it for inter-party exchange, adding file structure, naming, metadata, and security rules specific to Operator–Channel Partner usage export.

Each ExportJob instance executes a single data extraction based on pre-agreed parameters defined in the Agreement mapped into the selection criteria expressed in the query (time window, status).

(can be planned for future version) When processing completes, an ExportJobCreateEvent notifies the partner that data is available for download.

TM Forum references

TMF Asset	Purpose
TMF630 REST Guidelines	Defines the common exportJob resource
TMF635 Product Usage Management API	Defines the ProductUsage resource exported by the job and the common ExportJob resource

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TMF Asset	Purpose
TMF767 Product Usage Specification Management API	Defines the ProductUsageSpecification resource
SID ABEs: Product Usage, Product Usage Specification, Usage Export	Semantic alignment.

Purpose in TMF937

In TMF937, the ExportJob is the entity allowing delivering a list of finalized ProductUsage across organizational boundaries. It provides asynchronous, secure, and verifiable data exchange, supporting both Marketplace Aggregator and Wholesale models.

TMF937 exposes in a read-only mode pre-configured exportJobs. The creation and update operations of the exportJob resource are not managed via this API.

An ExportJob:

- Selects eligible ProductUsage records (status = rated or notRated).
- Packages them according to agreed parameters (for instance first productUsage creationDate and last creationDate).
- Generates a downloadable ZIP file containing usage data, metadata, and an optional signature.
- Notifies the Channel Partner upon completion.

Managed Entity and Task Resource Models

OpenGatewayExportJob resource

The base class for an ExportJob in the GSMA Open Gateway context, which extends the generic ExportJob class.

Resource model

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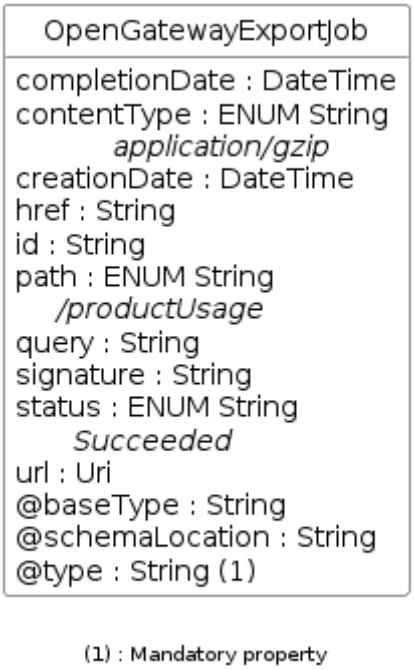


Figure 1 - OpenGatewayExportJob

Field descriptions

OpenGatewayExportJob fields

completionDate	A DateTime. Date on which the job was completed. This attribute exists only when status is Succeeded.
contentType	A String. The format of the export file. In Open Gateway context it is always application/gzip. ENUMERATED with values: * application/gzip
creationDate	A DateTime. Date on which the job was created.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
path	A String. JSON Pointer relative to the API root node that points to the resource acting as the source for streaming content to the export destination specified by the ExportJob. Set to /productUsage in Open Gateway context for this API. ENUMERATED with values: * /productUsage
query	A String. Used to scope the exported data. Identical to GET filter construct using the API node defined by path as base. In Open Gateway context, contains the query parameters used to filter ProductUsage records for export, in particular start and end completionDate.
signature	A String. JWS (JSON Web Signature) signature of the exported zip file. Used to verify the authenticity and integrity of the exported data.

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status	A String. Status of the export job. In this context can only be Succeeded. There is no requirement to share non Succeeded ExportJob. ENUMERATED with values: * Succeeded
url	A Uri. URL of the file containing exported data.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Json representation sample(s)

We provide below a JSON representation as example of the 'OpenGatewayExportJob' resource object.


```
{
  "signature":
"eyJhbGciOiJSUzI1NiIsDQoieDVjIjogWyANCiJNSUlCSWpBTkJna3Foa2lHOXcwQkFRRUZBQU9DQVE4QU1
JSUJDZ0tDQVFFQTF0UHZ3eWRQUzJOMVI4VVcyM1hzQzJ3eXdqSEdTUUZlbGVkTmNhSHV5aF95NHB6SjhvbDd
OZHp2bzRQNmF6dGNjRnZDY3VXekd0WHY3TmhZazJKd24xWWtmdnVBTS1qUTV6MjlYS2F6YThZcEtxUk5USTB
vRWZSRndVQ3BtVF9KYk5kMnQydUdXMzNGVWI3MHI1Z3AwRmw1cUNOX2g1cU03Y1g2RDVUM0hWeXRXbFc0ZlV
VTkhpVG9nZnJHanNjS3pDb0ZOdmtFeFRTUTdocW5ReGdzY1BBZ0Y4VGFmNkd2TzhBQUFxa2I1X25JYmNxZFU
xQlZVSFMyVVdISHVNR2JLMkxUVXE5bEFybGZDY2Q5NlItVWtNQXBGbFZQLTRraTJReWROanFhU2FrU1ZWbmt
YU3M5bmZQVnlMZWdyd2Yxcl8tZ19QdzlDQ2tjUGgyT3JXM2g2d0lEQVFBQiINCl0NCn0..B_YtCrSnlsEiqR
AcwO1-S0YHp486I8I4te9D-QzF2y0nKrt-
v6l9pzp9TngVrbkJ42shoUyiCsT5bet2HsQLJiBT1bQWOlf3unVEx17Z2sTGr1tZml4M2KWUoNvdVuqyGB8c
o43taSGnQ05zeRfs-C7JD_Z_cx6byhCbyhr-
bmTiF1St7lUYbxVDwpCMl7gAf4TQdM48inmQGRKjjEBHXJKhaBXcADaTiyDlEc1K8eWNGoxhxaJWJQx0yDkb
p2YhxNKBXhWDTBhvsRIIwCSFFFT9LjiXo8ouaSrDA-
2wrZz2NEarl4cc4nZTUL8ducMCJYDUmFnHVF6jb03G4MrVjQ",
  "@type": "OpenGatewayExportJob",
  "query": "status=rated,notRated&completionDate.gte=2025-02-
07T00:00:00.000Z&completionDate.lt=2025-02-08T00:00:00.000Z",
  "completionDate": "2025-02-08T00:02:35Z",
  "id": "8e8ce309-835a-4f4d-8ca2-0b33467e8fac",
  "href": "https://mycsp.com/tmf-api/openGatewayProductUsage/v5/exportJob/8e8ce309-
835a-4f4d-8ca2-0b33467e8fac",
  "creationDate": "2025-02-08T00:01:00Z",
  "contentType": "application/gzip",
  "url":
"sftp://mycsp.com/openGateway/openGatewayProductUsage/CINOP001CP00120250207T000000.0
00Z-20250208T000000.000Z.zip",
  "status": "Succeeded"
}
```


Key attributes kept from TMF635

Attribute	Description	Comment
id	Unique identifier of the export job.	—
href	URI of the resource.	—

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Attribute	Description	Comment
path	Path of the resource to export (e.g. /productUsage).	Fixed value /productUsage.
query	Filter expression defining data selection.	Must include status=rated,notRated and date interval using productUsage.completionDate.
contentType	Format of exported file.	Typically application/gzip.
url	Location of exported file (if status=Succeeded).	SFTP endpoint secured with mutual TLS.
status	Current processing status of the export job.	Standard lifecycle (see below). For this context, only ExportJob with status = Succeeded should be exposed.
creationDate	Timestamp when the job was initiated.	—
completionDate	Timestamp when the job completed (optional).	—

Additional attributes from base schema

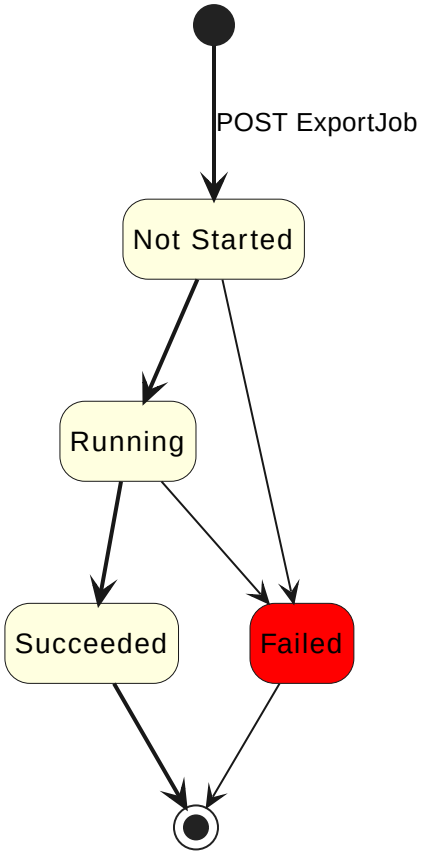
Attribute	Description	Comment
signature	string. Optional signature passed by value in JWS (JSON Web Signature) format.	TMF937 extension for data integrity.

Status Lifecycle (ExportJob.status)

status represents the processing status of the export job itself (distinct from ProductUsage.status)

Status	Description	(can be planned for future version) Event Trigger
Not Started	Job has been created and queued for processing.	Triggers ExportJobCreateEvent
Running	Data selection and packaging ongoing.	
Succeeded	Export completed successfully; file ready for download.	Triggers ExportJobStateChangeEvent
Failed	Processing failed; no valid output produced.	Optional ExportJobFailureEvent

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Notification Resource Models

In this API specification version there are no notifications defined in order to keep the focus on the basic requirements. This can be added in future backward compatible versions if required.

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API OPERATIONS

Remember the following Uniform Contract:

Operation on Entities	Uniform API Operation	Description
Query Entities	GET Resource	GET must be used to retrieve a representation of a resource.
Create Entity	POST Resource	POST must be used to create a new resource
Partial Update of an Entity	PATCH Resource	PATCH must be used to partially update a resource
Remove an Entity	DELETE Resource	DELETE must be used to remove a resource
Execute an Action on an Entity	POST on TASK Resource	POST must be used to execute Task Resources
Other Request Methods	POST on TASK Resource	GET and POST must not be used to tunnel other request methods.

Filtering and attribute selection rules are described in the TMF REST Design Guidelines.

Notifications are also described in a subsequent section.

Operations on OpenGatewayExportJob

List or find exportJob objects

`GET /exportJob?fields=…`

Description

This operation list OpenGatewayExportJob entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

List all exportJob with status Succeeded

Request

```
GET /exportJob?status=Succeeded
Content-Type: application/json
```

Response

```
200
[
```

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```
    {
        "id": "8e8ce309-835a-4f4d-8ca2-0b33467e8fac",
        "href": "https://mycsp.com/tmf-
api/openGatewayProductUsage/v5/exportJob/8e8ce309-835a-4f4d-8ca2-0b33467e8fac",
        "@type": "OpenGatewayExportJob",
        "path": "/productUsage",
        "query": "status=rated,notRated&completionDate.gte=2025-02-
07T00:00:00.000Z&completionDate.lt=2025-02-08T00:00:00.000Z",
        "creationDate": "2025-02-08T00:01:00Z",
        "completionDate": "2025-02-08T00:02:35Z",
        "contentType": "application/gzip",
        "url":
"sftp://mycsp.com/openGateway/openGatewayProductUsage/CINOP001CP00120250207T000000.0
00Z-20250208T000000.000Z.zip",
        "signature":
"eyJhbGciOiJSUzI1NiIsDQoieDVjIjogWyANCiJNSUlCSWpBTkJna3Foa2lHOXcwQkFRUZBQU9DQVE4QU1
JSUJDZ0tDQVFFQTF0UHZ3eWRQUzJOMVI4VVcyM1hzQzJ3eXdqSEdTUUZlbGVkTmNhSHV5aF95NHB6SjhvbDd
OZHp2bzRQNmF6dGNjRnZDY3VXekd0WHY3TmhZazJKd24xWWtmdnVBTS1qUTV6MjlYS2F6YThZcEtxUk5USTB
vRWZSRndVQ3BtVF9KYk5kMnQydUdXMzNGVWI3MHI1Z3AwRmw1cUNOX2g1cU03Y1g2RDVUM0hWeXRXbFc0ZlV
VTkhpVG9nZnJHanNjS3pDb0ZOdmtFeFRTUTdocW5ReGdzY1BBZ0Y4VGFmNkd2TzhBQUFxa2I1X25JYmNxZFU
xQlZVSFMyVVdISHVNR2JLMkxUVXE5bEFybGZDY2Q5NlItVWtNQXBGbFZQLTRraTJReWROanFhU2FrU1ZWbmt
YU3M5bmZQVnlMZWdyd2Yxcl8tZ19QdzlDQ2tjUGgyT3JXM2g2d0lEQVFBQiINCl0NCn0..B_YtCrSnlsEiqR
AcwO1-S0YHp486I8I4te9D-QzF2y0nKrt-
v6l9pzp9TngVrbkJ42shoUyiCsT5bet2HsQLJiBT1bQWOlf3unVEx17Z2sTGr1tZml4M2KWUoNvdVuqyGB8c
o43taSGnQ05zeRfs-C7JD_Z_cx6byhCbyhr-
bmTiF1St7lUYbxVDwpCMl7gAf4TQdM48inmQGRKjjEBHXJKhaBXcADaTiyDlEc1K8eWNGoxhxaJWJQx0yDkb
p2YhxNKBXhWDTBhvsRIIwCSFFFT9LjiXo8ouaSrDA-
2wrZz2NEarl4cc4nZTUL8ducMCJYDUmFnHVF6jb03G4MrVjQ",
        "status": "Succeeded"
    },
    {
        "id": "7a743f1c-daa2-4876-a439-c7f37af85204",
        "href": "https://mycsp.com/tmf-
api/openGatewayProductUsage/v5/exportJob/7a743f1c-daa2-4876-a439-c7f37af85204",
        "@type": "OpenGatewayExportJob",
        "path": "/productUsage",
        "query": "status=rated,notRated&completionDate.gte=2025-02-
08T00:00:00.000Z&completionDate.lt=2025-02-09T00:00:00.000Z",
        "creationDate": "2025-02-09T00:01:31Z",
        "completionDate": "2025-02-09T00:02:32Z",
        "contentType": "application/gzip",
        "url":
"sftp://mycsp.com/openGateway/openGatewayProductUsage/CINOP001CP00120250208T000000.0
00Z-20250209T000000.000Z.zip",
        "signature":
"eyJhbGciOiJSUzI1NiIsDQoieDVjIjogWyANCiJNSUlCSWpBTkJna3Foa2lHOXcwQkFRUZBQU9DQVE4QU1
JSUJDZ0tDQVFFQTF0UHZ3eWRQUzJOMVI4VVcyM1hzQzJ3eXdqSEdTUUZlbGVkTmNhSHV5aF95NHB6SjhvbDd
OZHp2bzRQNmF6dGNjRnZDY3VXekd0WHY3TmhZazJKd24xWWtmdnVBTS1qUTV6MjlYS2F6YThZcEtxUk5USTB
vRWZSRndVQ3BtVF9KYk5kMnQydUdXMzNGVWI3MHI1Z3AwRmw1cUNOX2g1cU03Y1g2RDVUM0hWeXRXbFc0ZlV
VTkhpVG9nZnJHanNjS3pDb0ZOdmtFeFRTUTdocW5ReGdzY1BBZ0Y4VGFmNkd2TzhBQUFxa2I1X25JYmNxZFU
xQlZVSFMyVVdISHVNR2JLMkxUVXE5bEFybGZDY2Q5NlItVWtNQXBGbFZQLTRraTJReWROanFhU2FrU1ZWbmt
YU3M5bmZQVnlMZWdyd2Yxcl8tZ19QdzlDQ2tjUGgyT3JXM2g2d0lEQVFBQiINCl0NCn0..B_YtCrSnlsEiqR
AcwO1-S0YHp486I8I4te9D-QzF2y0nKrt-
v6l9pzp9TngVrbkJ42shoUyiCsT5bet2HsQLJiBT1bQWOlf3unVEx17Z2sTGr1tZml4M2KWUoNvdVuqyGB8c
o43taSGnQ05zeRfs-C7JD_Z_cx6byhCbyhr-
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    },
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```


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    }
]
```


Retrieves a exportJob by ID

`GET /exportJob/{id}?fields=…`

Description

This operation retrieves an OpenGatewayExportJob entity. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Retrieve a single exportJob

Request

```
GET /exportJob/8e8ce309-835a-4f4d-8ca2-0b33467e8fac
Content-Type: application/json
```

Response


```
200
{
      "id": "8e8ce309-835a-4f4d-8ca2-0b33467e8fac",
      "href": "https://mycsp.com/tmf-
api/openGatewayProductUsage/v5/exportJob/8e8ce309-835a-4f4d-8ca2-0b33467e8fac",
      "@type": "OpenGatewayExportJob",
      "path": "/productUsage",
      "query": "status=rated,notRated&completionDate.gte=2025-02-
07T00:00:00.000Z&completionDate.lt=2025-02-08T00:00:00.000Z",
      "creationDate": "2025-02-08T00:01:00Z",
      "completionDate": "2025-02-08T00:02:35Z",
      "contentType": "application/gzip",
```


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      "url":
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00Z-20250208T000000.000Z.zip",
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Acknowledgements

Release History

Release Number	Date	Release led by:	Description
Version 5.0.0-beta1	January 2026	Elisabeth Andersson (MATRIXX)	

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